1.6.4 ▶ Damped Oscillations

In section 1.6.1, we looked at an idealistic version of a simple harmonic oscillator. In real life, though, oscillators will have some kind of resistive force acting on them. Consider a spring oscillator with mass m, spring constant k, and in a medium that provides a velocity-dependent resistive force of magnitude $F_d = bv$. Writing Newton's Second law for this system gives

$$m\ddot{x} = -b\dot{x} - kx$$

$$m\ddot{x} + b\dot{x} + kx = 0$$

Just like with the simple harmonic oscillator, this is a **homogeneous second order linear ordinary differential equation**. Thus, we employ the same method to solve it. Guessing a solution in the form of $x(t) = Ae^{\alpha t}$ and plugging in gives

$$m\alpha^2 + b\alpha + k = 0 \tag{1.94}$$

using the quadratic formula, we have

$$\begin{aligned}\alpha &= -\frac{b}{2m} \pm \frac{\sqrt{b^2 - 4mk}}{2m} \\ &= -\frac{b}{2m} \pm \sqrt{\frac{b^2}{4m^2} - \frac{k}{m}} \\ &= -\frac{1}{\tau} \pm \sqrt{\frac{1}{\tau^2} - \boldsymbol{\omega}_0^2}. \end{aligned} \tag{1.95}$$

Where $\boldsymbol{\omega}_0 = \sqrt{\frac{k}{m}}$, and $\tau = \frac{2m}{b}$. Now, we can identify a few cases for the relative sizes of τ and $\boldsymbol{\omega}_0$.

Case 1: $\frac{1}{\tau^2} - \boldsymbol{\omega}_0^2 > 0$
In this case, the solution to the differential equation is

$$x(t) = Ae^{t/\tau}\left(e^{\sqrt{\frac{1}{\tau^2} - \boldsymbol{\omega}_0^2}\cdot t} + e^{-\sqrt{\frac{1}{\tau^2} - \boldsymbol{\omega}_0^2}\cdot t}\right) \tag{1.96}$$

This case is known as overdamping, and it indicates that the oscillator goes back to the equilibrium position before completing one full oscillation.

Case 2: $\frac{1}{\tau^2} - \boldsymbol{\omega}_0^2 = 0$
In this case, the solution is simply

$$x(t) = Ae^{-t/\tau} \tag{1.97}$$

Again, the oscillator returns to the equilibrium position before completing a full oscillation, but this time it reaches equilibrium in the fastest time possible. Thus, this case is known as critical damping.

Case 3: $\frac{1}{\tau^2} - \boldsymbol{\omega}_0^2 < 0$
In this case, we get an imaginary number in the exponent, so we can use Euler's identity to simplify, similar to the case for SHM. Thus, the solution to the differential equation is

$$x(t) = Ae^{-t/\tau}\cos(\boldsymbol{\omega}'t + \phi) \tag{1.98}$$

where $\boldsymbol{\omega}' = \sqrt{\boldsymbol{\omega}_0^2 - \frac{1}{\tau^2}}$ and ϕ is a constant. This case is known as underdamping, and the oscillator completes full oscillations, with its amplitude decaying to zero at infinity. Note that the frequency of the oscillation in this case doesn't actually change, and only the amplitude does. This is because the resistive force decreases the total energy of the system, which is dependent on the amplitude of the motion, and not the frequency.